SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.4: Linear Motion — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 3.4: Linear Motion
Driving Question	DRIVING QUESTION: How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line — and why does direction always matter?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Distance vs. Displacement: Why Direction Matters from the Start**	Students observed a demonstration (or video) of a matatu travelling from Nairobi CBD to Westlands and back. They noted the total path covered and the start-to-finish straight-line separation. Teacher cold-called at least 3 students to describe what they noticed about the two measures. A wait time of 10 seconds was used after each prompt to encourage deeper thinking.	Distance is a scalar quantity — it measures the total length of the path travelled, regardless of direction. Displacement is a vector quantity — it measures the shortest straight-line distance from the starting point to the finishing point, including direction. When an object returns to its starting point, displacement = 0 even though distance > 0. Pedagogical note: Stress that this distinction underpins every lesson in the unit; students who conflate distance and displacement will struggle with all subsequent velocity and graph work.	The matatu scenario was used as an anchor: the matatu covers 5 km to Westlands and 5 km back, so distance = 10 km but displacement = 0 km. Students were asked to explain in writing why a Kisumu fisherman who sails 3 km east then 3 km west has zero displacement but a tired body. Teacher noted that direction is encoded in displacement using positive/negative signs or compass bearings — connecting forward to the driving question's phrase 'direction always matters.'
Lesson 2 **Speed vs. Velocity: Scalars, Vectors, and the Formula v = s ÷ t**	Students observed two scenarios side-by-side: (a) a sprinter running 100 m along a straight track at Nyayo Stadium and (b) a boda-boda rider completing a circular route around a Nairobi estate. Data cards showed time and path length for each. Teacher asked students to predict — before any instruction — which rider had greater speed and which had greater velocity, allowing 10 seconds of think time before cold-calling 4 students.	Speed = distance ÷ time and is a scalar (magnitude only). Velocity = displacement ÷ time and is a vector (magnitude AND direction). Two objects can have the same speed but different velocities if they are moving in different directions. The SI unit for both is metres per second (m/s). Pedagogical note: Use the boda-boda example deliberately — if the rider returns to the start in 60 s, speed > 0 but velocity = 0, providing a concrete hook for why direction cannot be ignored.	Students applied v = s ÷ t to calculate the sprinter's velocity (displacement = 100 m, time = 10 s, v = 10 m/s north) and the boda-boda rider's average velocity (displacement ≈ 0, so average velocity ≈ 0 m/s). They explained the apparent paradox — busy motion yet near-zero velocity — by linking back to the displacement definition from Lesson 1. Teacher reinforced the driving question: mathematics (the formula and the sign of displacement) is the tool that captures direction.

Lesson 3 **Velocity, Speed & Acceleration: Illustrations + Calculations**	Students observed illustrated scenarios: a matatu accelerating from a bus stop along Thika Road, a cyclist braking to a halt at a Kisumu roundabout, and a Rift Valley Express bus cruising at constant velocity on the Nakuru–Nairobi highway. For each scenario, students recorded whether velocity was changing and, if so, in what direction. Teacher provided 10 seconds of wait time before cold-calling 3 students per scenario to justify their reasoning.	Velocity is the rate of change of displacement with time and is a vector quantity. Acceleration is the rate of change of velocity with time: $a = (v - u) \div t$, where u = initial velocity and v = final velocity. Acceleration is also a vector — its sign indicates whether the object is speeding up in the positive direction (positive a) or slowing down / moving in the negative direction (negative a, deceleration). Uniform acceleration means the velocity changes by equal amounts in equal time intervals. Pedagogical note: Emphasise that 'deceleration' is simply negative acceleration; students should practise assigning signs consistently from a defined positive direction.	Students calculated acceleration for each scenario: e.g., the matatu accelerates from 0 to 20 m/s in 10 s → $a = (20-0)\div 10 = 2$ m/s². The cyclist decelerates from 8 m/s to 0 in 4 s → $a = (0-8)\div 4 = -2$ m/s². They explained that the negative sign for the cyclist's acceleration does NOT mean slow motion — it means the acceleration vector points opposite to the direction of travel, logically connecting deceleration to the sign convention established in Lesson 1. Link to driving question: mathematics (the formula and its sign) fully describes how motion is changing.
Lesson 4 **Velocity and Acceleration Problems — Practice & Sensemaking**	Students worked through a set of structured practice problems drawn from Kenyan contexts: a Nairobi–Mombasa SGR train reaching 120 km/h from rest over 5 minutes, a hawker accelerating on a bicycle down a Kericho hill, and a school sprinter at a county athletics meet. Students first predicted the sign of acceleration for each scenario before calculating. Teacher circulated and used cold-calling (minimum 5 students) to surface misconceptions, with 10 seconds of wait time per cold-call.	Velocity = displacement ÷ time and is a vector — its sign encodes direction. Acceleration = $(v - u) \div t$ and is also a vector — positive acceleration means gaining speed in the chosen positive direction; negative acceleration (deceleration) means losing speed or gaining speed in the negative direction. Units must be consistent (convert km/h to m/s where necessary: divide by 3.6). Pedagogical note: The most common error at this stage is ignoring sign conventions; insist students define a positive direction at the top of every problem before substituting values.	Students compared their predicted signs against calculated values and explained any discrepancies. For the SGR train: $u = 0$, $v = 120$ km/h = 33.3 m/s, $t = 300$ s → $a = 0.11$ m/s² (positive, as expected). For a cyclist braking: $u = 10$ m/s, $v = 0$, $t = 5$ s → $a = -2$ m/s². Students articulated that the formula $a = (v - u) \div t$ is the mathematical tool that answers the driving question — it quantifies HOW MUCH and IN WHAT DIRECTION motion is changing.
Lesson 5 **Drawing Displacement–Time Graphs: Reading Motion from a Line**	Students were given a table of displacement–time data for a Kisumu fisherman walking from the lakeshore market to his boat and back. They plotted the points on grid paper (or a digital tool) and observed the shape of each segment. Teacher asked students to predict the meaning of a steep line vs. a shallow line vs. a horizontal line before drawing, allowing 10 seconds of wait time and cold-calling 4 students. Students annotated their graphs with arrows showing direction of motion.	The gradient (slope) of a displacement–time graph equals the velocity of the object: gradient = $\Delta s \div \Delta t$. A positive gradient indicates motion in the positive direction (away from the reference point). A negative gradient indicates motion in the negative direction (returning toward the reference point). A horizontal line (gradient = 0) means the object is stationary — displacement is not changing. A steeper line means greater speed. Pedagogical note: Insist students always label axes with units and scale; unlabelled graphs are a persistent exam error.	Students calculated the gradient of each segment of the fisherman's graph and matched each gradient value to a velocity. For example, Segment A: $\Delta s = 600$ m, $\Delta t = 5$ min → gradient = 2 m/s (walking to boat); Segment B: gradient = 0 (loading fish, stationary); Segment C: negative gradient (returning). Students explained how the graph tells the complete story of the fisherman's journey — speed, direction, and rest — connecting to the driving question: a single graph mathematically encodes all aspects of linear motion.

Lesson 6 **Interpreting Displacement–Time Graphs**	Students were presented with a pre-drawn displacement–time graph (without a narrative) showing multiple segments of varying gradients, including a negative segment and a flat section, representing an unidentified moving object in a Nairobi context. Working in pairs, they described in words what the object was doing in each phase. Teacher cold-called 5 pairs (10-second wait time each) before revealing the context: a delivery rider collecting and dropping off parcels along Ngong Road.	The gradient (slope) of a displacement–time graph equals the velocity of the object for that segment. A steep positive gradient = high positive velocity (fast motion away from origin). A gentle positive gradient = low positive velocity (slow motion away from origin). A zero gradient = zero velocity (object is at rest). A negative gradient = negative velocity (object moving back toward origin). The graph is a complete mathematical description of the object's motion history. Pedagogical note: A common misconception is that a 'high point' on the graph means the object is moving fast — clarify that height shows position, while steepness shows speed.	Students calculated the velocity for each segment by computing $\Delta s \div \Delta t$ using coordinates read from the graph. They then wrote a coherent narrative of the delivery rider's journey, including how far from base each stop was and in what direction the rider was moving at each stage. They explicitly connected gradient = velocity back to the formula $v = s \div t$ from Lesson 2, showing these are the same mathematical idea expressed differently. Link to driving question: reading a graph is reading motion — direction is encoded in the sign of the gradient.
Lesson 7 **Drawing Velocity–Time Graphs: Gradients, Acceleration, and Kipchoge's Splits**	Students were given a table of Eliud Kipchoge's approximate velocity data at different time splits during a marathon, segmented into: warming up (increasing velocity), cruising (near-constant velocity), and final sprint (rapidly increasing velocity). Students plotted the velocity–time graph and predicted — before instruction — what the gradient of each segment would tell them. Teacher used 10 seconds of wait time and cold-called 4 students to share predictions.	The gradient of a velocity–time graph equals the acceleration of the object: $a = \Delta v \div \Delta t$. A positive gradient means positive acceleration (speeding up in the positive direction). A zero gradient (horizontal line) means zero acceleration — the object moves at constant (uniform) velocity. A negative gradient means negative acceleration (deceleration — slowing down). The steeper the gradient, the greater the magnitude of acceleration. Pedagogical note: Contrast with displacement–time graphs carefully — on a v–t graph, a horizontal line ≠ stationary; it means constant velocity. This is a very high-frequency exam misconception.	Students calculated Kipchoge's acceleration during each phase. For example, warm-up phase: v increases from 3 m/s to 5.5 m/s over 600 s → $a = 2.5 \div 600 \approx 0.004$ m/s² (small, gradual). Cruising phase: $v \approx$ constant → $a \approx 0$. Final sprint: v increases sharply → steeper positive gradient. Students explained why Kipchoge's ability to maintain a near-zero acceleration (constant velocity) over most of the marathon is the signature of elite pacing, grounding the mathematics in a Kenyan world-record context. Link to driving question: the gradient of a v–t graph is the mathematical tool that reveals how motion is changing.
Lesson 8 **Interpreting Velocity–Time Graphs & Area Under the Graph**	Students were shown a pre-drawn velocity–time graph representing a Mombasa ferry crossing a harbour channel: accelerating from rest, cruising at constant velocity, then decelerating to dock. They were asked to predict (a) the acceleration during each phase and (b) what the shaded area under the graph might represent physically. Teacher used 10 seconds of wait time and cold-called at least 5 students before any explanation was given.	The gradient of a velocity–time graph equals acceleration ($\Delta v \div \Delta t$); a negative gradient indicates deceleration. The area under a velocity–time graph equals the displacement of the object. For a rectangular section: area = base × height = t × v. For a triangular section (uniform acceleration/deceleration): area = ½ × base × height = ½ × t × v. Total displacement = sum of all areas under the graph (taking areas below the time axis as negative displacement). Pedagogical note: Students frequently confuse gradient and area — explicitly contrast: gradient → acceleration;	Students calculated the total displacement of the ferry by computing the area of the trapezium (or sum of triangle + rectangle + triangle) under the graph. They verified the result using $s = ut + \frac{1}{2}at^2$ for the acceleration phase and compared. They then explained why direction matters: if the ferry reverses (v goes negative), that section's area is subtracted. Teacher connected this to Lesson 1 (displacement vs. distance): total area with signs = displacement; total area without signs = distance. Full link to driving question: area under a v–t graph is the mathematical

		area → displacement. Using different colours for each segment helps.	bridge between velocity graphs and displacement — completing the toolkit.
Lesson 9 **Relative Speed — Combining Velocities When Direction Matters**	Students observed two scenarios using toy cars (or a simulation/video): (a) two matatus travelling in the same direction along Uhuru Highway at different speeds, and (b) two buses approaching each other head-on between Nairobi and Nakuru. Students predicted which scenario produces a greater 'closing speed' and why. Teacher used 10 seconds of wait time and cold-called 4 students per scenario, probing the role of direction in each answer before any formal rules were given.	When two objects move in the same direction, relative speed = $\|v_1 - v_2\|$ (the difference of their speeds). When two objects move in opposite directions, relative speed = $v_1 + v_2$ (the sum of their speeds). Relative speed is always the speed of one object as observed by the other. Direction determines which formula applies — this is a direct consequence of velocity being a vector. Pedagogical note: Use a consistent sign convention (e.g., eastward = positive) and show algebraically why opposite directions yield a sum: if $v_1 = +30$ m/s and $v_2 = -25$ m/s, relative velocity = +30 − (−25) = +55 m/s. This algebraic derivation prevents rote memorisation of 'add or subtract' rules.	Students solved problems: (a) Two SGR trains on parallel tracks both heading Nairobi→Mombasa at 40 m/s and 35 m/s — relative speed = 5 m/s. (b) A Nairobi-bound bus at 25 m/s meets a Nakuru-bound bus at 20 m/s — relative speed = 45 m/s. They explained why scenario (b) feels more dangerous (objects close at 45 m/s instead of 5 m/s) and connected this to real-world road safety in Kenya. Link to driving question: relative speed is impossible to calculate correctly without knowing direction — reinforcing the unit's central theme.
Lesson 10 **Application & Cumulative Model Building: The Final Explanation**	Students reviewed their Driving Question Board entries from all previous lessons and identified which questions had been answered and which remained. They were then presented with a rich, multi-phase scenario: a matatu leaves Nairobi, accelerates uniformly, cruises, decelerates at a roundabout, reverses slightly, then continues to Thika — with full numerical data. Students predicted which mathematical tools from the unit they would need to fully describe this journey before beginning any calculations. Teacher cold-called 6 students (10-second wait time each) to name specific tools and justify their relevance.	The full mathematical toolkit for linear motion integrates: (1) scalar vs. vector distinction (distance/speed vs. displacement/velocity); (2) $v = s \div t$ for uniform velocity; (3) $a = (v - u) \div t$ for acceleration; (4) displacement–time graphs (gradient = velocity); (5) velocity–time graphs (gradient = acceleration; area = displacement); and (6) relative speed rules (same direction: subtract; opposite: add). No single tool is sufficient alone — the complete description of linear motion requires all of them working together. Pedagogical note: This lesson is the capstone; use it diagnostically to identify any remaining gaps before summative assessment.	Students constructed a final cumulative model — a annotated diagram and matching set of graphs — for the matatu scenario, labelling every phase with the correct mathematical description: displacement, velocity, acceleration, and graph shape. They explicitly answered the driving question in writing: 'Mathematics helps us describe linear motion through formulas for velocity and acceleration, displacement–time and velocity–time graphs, and relative speed rules. Direction always matters because velocity and acceleration are vectors — ignoring direction gives incomplete or incorrect answers, as shown by the difference between distance and displacement, speed and velocity, and the two relative-speed formulas.' Teacher collected written responses as a formative assessment artefact.